

ALFAA44821

## Nickel foil

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:** 镍箔, 0.1mm (0.004in) 厚  
**Product Description:** Nickel foil

**Cat No. :** 44821  
**CAS No** 7440-02-0  
**Molecular Formula** Ni

**Supplier** Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
Shore Road, Heysham  
Lancashire, LA3 2XY,  
United Kingdom  
Office Tel: +44 (0) 1524 850506  
Office Fax: +44 (0) 1524 850608

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Solid

**Appearance**  
No information available

**Odor**  
No information available

#### Emergency Overview

Suspected of causing cancer. May cause an allergic skin reaction. Causes damage to organs through prolonged or repeated exposure.

#### Classification of the substance or mixture

|                                                      |            |
|------------------------------------------------------|------------|
| Skin Sensitization                                   | Category 1 |
| Carcinogenicity                                      | Category 2 |
| Specific target organ toxicity - (repeated exposure) | Category 1 |

#### Label Elements



**Signal Word**

**Danger**

**Hazard Statements**

H351 - Suspected of causing cancer  
H317 - May cause an allergic skin reaction  
H372 - Causes damage to organs through prolonged or repeated exposure

**Precautionary Statements****Prevention**

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves

**Response**

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 - Store in a well-ventilated place

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

Suspected of causing cancer. May cause an allergic skin reaction. Causes damage to organs through prolonged or repeated exposure.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Is not likely mobile in the environment due its low water solubility. Spillage unlikely to penetrate soil.

**Other Hazards**

Toxicity to Soil Dwelling Organisms. This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component | CAS No    | Weight % |
|-----------|-----------|----------|
| Nickel    | 7440-02-0 | >95      |

**SECTION 4. FIRST AID MEASURES****General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Most important symptoms and effects**

May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION**

**Control Parameters**

| Component | China                    | Taiwan                   | Thailand                 | Hong Kong                  |
|-----------|--------------------------|--------------------------|--------------------------|----------------------------|
| Nickel    | TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> | TWA: 1.5 mg/m <sup>3</sup> |

| Component | ACGIH TLV                  | OSHA PEL                                                       | NIOSH                                                      | The United Kingdom                                                         | European Union |
|-----------|----------------------------|----------------------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------|----------------|
| Nickel    | TWA: 1.5 mg/m <sup>3</sup> | (Vacated) TWA: 1 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | IDLH: 10 mg/m <sup>3</sup><br>TWA: 0.015 mg/m <sup>3</sup> | STEL: 1.5 mg/m <sup>3</sup> 15 min<br>TWA: 0.5 mg/m <sup>3</sup> 8 hr Skin |                |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls****Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |
| Nitrile rubber |                                   |                 |             |                       |
| Neoprene       |                                   |                 |             |                       |
| PVC            |                                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

|                                 |                                                                                                                                                                                   |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hygiene Measures                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                            |
| Environmental exposure controls | Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained. |

**SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES**

|                                         |                                             |                                   |
|-----------------------------------------|---------------------------------------------|-----------------------------------|
| Appearance                              |                                             |                                   |
| Physical State                          | Solid                                       |                                   |
| Odor                                    | No information available                    |                                   |
| Odor Threshold                          | No data available                           |                                   |
| pH                                      | No information available                    |                                   |
| Melting Point/Range                     | 1455 °C / 2651 °F                           |                                   |
| Softening Point                         | No data available                           |                                   |
| Boiling Point/Range                     | 2732 - 2837 °C / 4949.6 - 5138.6 @ 760 mmHg |                                   |
|                                         | °F                                          |                                   |
| Flash Point                             | No information available                    | Method - No information available |
| Evaporation Rate                        | Not applicable                              | Solid                             |
| Flammability (solid,gas)                | No information available                    |                                   |
| Explosion Limits                        | No data available                           |                                   |
| Vapor Pressure                          | 1 mmHg @ 1810 °C                            |                                   |
| Vapor Density                           | Not applicable                              | Solid                             |
| Specific Gravity / Density              | No data available                           |                                   |
| Bulk Density                            | No data available                           |                                   |
| Water Solubility                        | Insoluble                                   |                                   |
| Solubility in other solvents            | No information available                    |                                   |
| Partition Coefficient (n-octanol/water) |                                             |                                   |
| Autoignition Temperature                | No data available                           |                                   |
| Decomposition Temperature               | No data available                           |                                   |
| Viscosity                               | Not applicable                              | Solid                             |
| Explosive Properties                    | No information available                    |                                   |
| Oxidizing Properties                    | No information available                    |                                   |
| Molecular Formula                       | Ni                                          |                                   |
| Molecular Weight                        | 58.7                                        |                                   |

**SECTION 10. STABILITY AND REACTIVITY**

|                          |                                                           |
|--------------------------|-----------------------------------------------------------|
| Stability                | Stable under normal conditions.                           |
| Hazardous Reactions      | None under normal processing.                             |
| Hazardous Polymerization | Hazardous polymerization does not occur.                  |
| Conditions to Avoid      | Incompatible products. Excess heat. Avoid dust formation. |
| Materials to avoid       | Strong oxidizing agents.                                  |

Hazardous Decomposition Products None under normal use conditions.

**SECTION 11. TOXICOLOGICAL INFORMATION****Product Information**

(a) acute toxicity;

**SAFETY DATA SHEET****Nickel foil**

| Component | LD50 Oral                 | LD50 Dermal | LC50 Inhalation              |
|-----------|---------------------------|-------------|------------------------------|
| Nickel    | LD50 > 9000 mg/kg ( Rat ) |             | LC50 > 10.2 mg/L ( Rat ) 1 h |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

Respiratory No data available  
Skin Category 1

No information available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; Category 2

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component | EU | UK | Germany | IARC     |
|-----------|----|----|---------|----------|
| Nickel    |    |    | Cat. 1  | Group 2B |

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; Category 1

Target Organs Kidney, Skin, Lungs, Blood.

(j) aspiration hazard; Not applicable  
Solid

**Symptoms / effects, both acute and delayed** Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

## SECTION 12. ECOLOGICAL INFORMATION

### Ecotoxicity effects

The product contains following substances which are hazardous for the environment.  
Contains a substance which is: Very toxic to aquatic organisms.

| Component | Freshwater Fish                                                                                                                                     | Water Flea          | Freshwater Algae                            | Microtox |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------|----------|
| Nickel    | LC50: > 100 mg/L, 96h (Brachydanio rerio)<br>LC50: = 1.3 mg/L, 96h semi-static (Cyprinus carpio)<br>LC50: = 10.4 mg/L, 96h static (Cyprinus carpio) | EC50 = 510 µg/L 96h | EC50 = 0.1 mg/L 72h<br>EC50 = 0.18 mg/L 72h |          |

### Persistence and Degradability

Persistence  
Degradability  
Degradation in sewage treatment plant

Insoluble in water.  
Not relevant for inorganic substances.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

|                                        |                                                                                                          |
|----------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Bioaccumulative Potential</b>       | May have some potential to bioaccumulate                                                                 |
| <b>Mobility in soil</b>                | Spillage unlikely to penetrate soil Is not likely mobile in the environment due its low water solubility |
| <b>Endocrine Disruptor Information</b> | This product does not contain any known or suspected endocrine disruptors                                |
| <b>Persistent Organic Pollutant</b>    | This product does not contain any known or suspected substance                                           |
| <b>Ozone Depletion Potential</b>       | This product does not contain any known or suspected substance                                           |

**SECTION 13. DISPOSAL CONSIDERATIONS**

|                                            |                                                                                                                                                                        |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations. |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                              |
| <b>Other Information</b>                   | Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.                   |

**SECTION 14. TRANSPORT INFORMATION**

**Road and Rail Transport** Not Regulated

**IMDG/IMO** Not regulated

**IATA** Not regulated

**Special Precautions for User** No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-----------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Nickel    | -                                                   | -                                       | X    | X     | 231-111-4 | X    | X   | X     | X    |      | X    | KE-25818 |

**National Regulations****SECTION 16. OTHER INFORMATION**

## Nickel foil

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|                         |                                                    |
|-------------------------|----------------------------------------------------|
| <b>Prepared By</b>      | Health, Safety and Environmental Department        |
| <b>Creation Date</b>    | 11-Feb-2010                                        |
| <b>Revision Date</b>    | 29-Apr-2024                                        |
| <b>Revision Summary</b> | New emergency telephone response service provider. |

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**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**